MODULE 1 · LECTURE 20

Turing Test and Control Strategies

Turing's imitation game, his prediction and the objections he answered, the main criticisms and variants of the test; then how an AI program solves problems as a search through states, and the control strategies that decide what it tries next.

Learning objectives

After this lecture you should be able to:

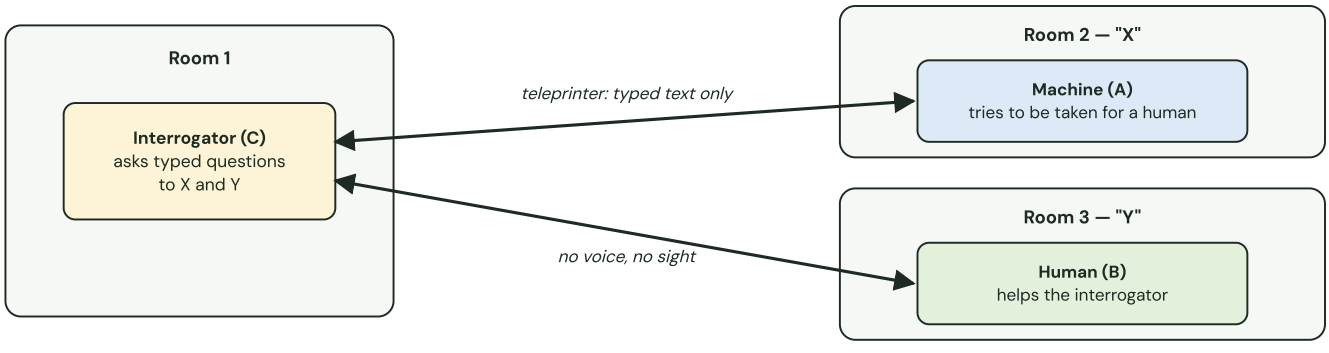
1. Describe the Turing test and the question it was designed to replace.
2. State Turing's prediction and summarise the objections he answered.
3. Explain criticisms and variants of the test: the Chinese room, ELIZA and CAPTCHA.
4. Represent a problem as a state space with an initial state, operators and a goal test.
5. Describe a production system and the requirements of a good control strategy.
6. Classify control strategies and trace breadth-first and depth-first control on the water-jug problem.

Scope: Lecture 19 introduced AI and mentioned the Turing test as the "acting humanly" approach. This lecture examines the test and then begins problem solving by search. Breadth-first and depth-first search are treated in detail in Lecture 21, and heuristic search in Lecture 22.

Part A — The Turing test

1. The imitation game

In 1950 Alan Turing considered the question "Can machines think?" and, finding it too vague, replaced it with a game (Turing, 1950). In the original **imitation game** an interrogator (C) questions a man (A) and a woman (B), who are out of sight, and must decide which is which. To stop tones of voice helping, the answers are typewritten — ideally through a teleprinter between the rooms. Turing then asked what would happen when a **machine takes the part of A**: would the interrogator be wrong as often as before?



After questioning, C must say which of X and Y is the machine. If C is often wrong, the machine plays the game well.

Figure 1. The imitation game with a machine (the Turing test). Original diagram.

Turing made a cautious forecast: "I believe that in about fifty years' time it will be possible to programme computers, with a storage capacity of about 10^9, to make them play the imitation game so well that an

average interrogator will not have more than 70 per cent. chance of making the right identification after five minutes of questioning" (Turing, 1950, p. 442).

> **Worked example 1 — reading a Turing-test result** (illustrative)
> Thirty judges each hold a five-minute conversation and 20 identify the machine correctly: success rate = 20 ÷ 30 = 66.7%, below Turing's 70% mark.
> But the standard error is √(0.667 × 0.333 ÷ 30) = 0.086, so an approximate 95% interval is 66.7% ± 1.96 × 8.6% = 49.8% to 83.5%.
> The interval includes 70%, so 30 judges cannot show whether the machine really meets the mark — a reminder (Lecture 11) that test results need enough trials.

2. Objections Turing answered

Turing listed and answered nine objections to the idea that machines could think (Turing, 1950):

Objection	Its claim	Turing's reply (in brief)
1. Theological	Thinking belongs to an immortal soul given only to humans	This limits divine power; theological arguments have often been wrong before
2. "Heads in the sand"	The consequences would be too dreadful, so let us hope machines cannot think	Not substantial enough to need refutation — consolation would be more appropriate
3. Mathematical	Gödel's theorem and related results show limits to what machines can do	Humans also make mistakes and have limits; a particular machine's limits say little
4. Consciousness	A machine cannot feel or know that it has written a sonnet	We judge other people's minds by their behaviour too; otherwise we fall into solipsism
5. Various disabilities	A machine can never be kind, learn from experience, fall in love, enjoy strawberries…	These are unproved claims, often based on the limited machines seen so far
6. Lady Lovelace	A machine can only do what we know how to order it to do; it can never take us by surprise	"Machines take me by surprise with great frequency"; and learning machines may do new things
7. Continuity of the nervous system	The brain is not a discrete-state machine	A digital computer can imitate a continuous machine closely enough in the game
8. Informality of behaviour	No set of rules can describe what a person would do in every situation	We cannot be sure there are no laws governing behaviour
9. Extra-sensory perception	Telepathy and similar abilities would give humans an advantage	A "telepathy-proof room" would be needed

3. Criticisms and variants of the test

- **The Chinese room** — Searle (1980) argued that running a program is never by itself sufficient for understanding: a person who follows rules for manipulating Chinese symbols could give correct answers in Chinese without understanding a word. Behaving intelligently, on this view, is not the same as thinking.
- **ELIZA** — Weizenbaum's 1966 program produced conversation by spotting keywords and applying decomposition and reassembly rules, with no understanding of meaning (Weizenbaum, 1966). It

shows how easily fluent text can be mistaken for understanding.

- **CAPTCHA, a reverse Turing test** — websites use problems that people solve easily but machines find hard to tell humans and programs apart, an idea built on hard AI problems (von Ahn et al., 2003).
- **Total Turing test** — some researchers ask that the machine also perceive and manipulate objects, adding computer vision and robotics to language.

Modern chatbots can hold fluent conversations, but for agricultural services the useful questions are different from Turing's: is the advice accurate and locally valid, does the system admit uncertainty, and are users told they are talking to an AI (Lecture 19)?

Part B — Control strategies

4. Problem solving as a search through states

Many AI problems can be stated as finding a sequence of actions that turns a starting situation into a goal (Russell & Norvig, 2020). A **state-space** formulation has:

- **States** — descriptions of the situation, e.g. (litres in the 4-L can, litres in the 3-L can).
- **Initial state** — where the search begins, e.g. (0, 0).
- **Operators (actions, rules)** — ways of changing one state into another: fill, empty or pour a can.
- **Goal test** — whether a state is a solution, e.g. 2 litres in the 4-L can.
- **Path cost** — the cost of a sequence of actions, often the number of steps.

Production systems

A **production system** organises the search as a **global database** (the current state), a set of **production rules** (IF condition THEN action) and a **control system** that decides which applicable rule to apply next and when to stop (Nilsson, 1980).

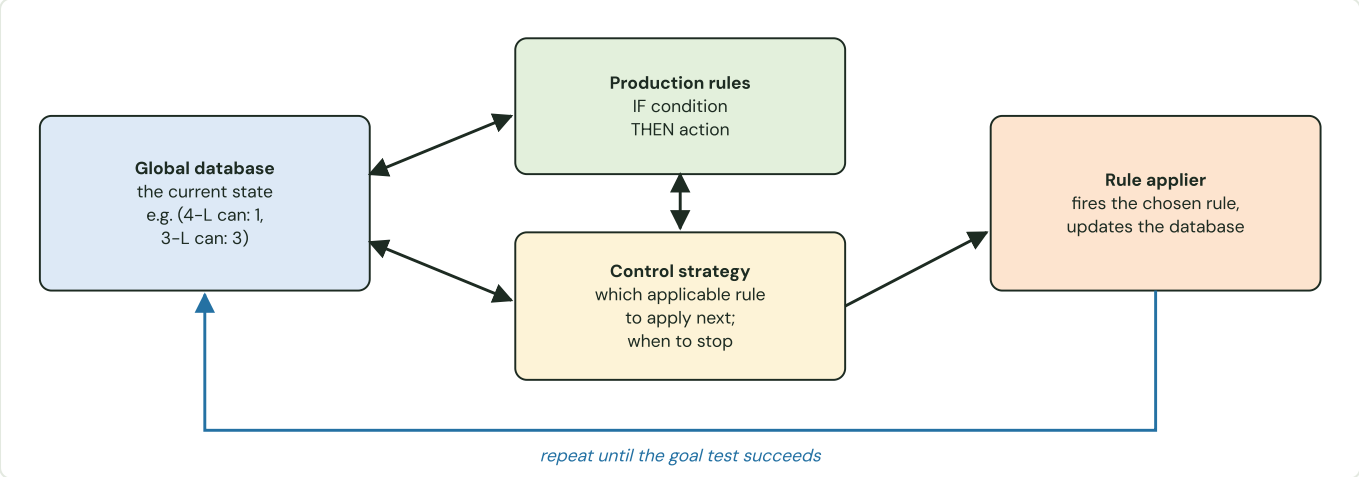


Figure 2. A production system. Original diagram.

5. Control strategies

Usually several rules apply at each step, and the **control strategy** decides which to try. A good control strategy must **cause motion** — every step should move the search on, not repeat it — and must be **systematic**, exploring the alternatives in an orderly way without wandering back to states already tried (Rich et al., 2009).

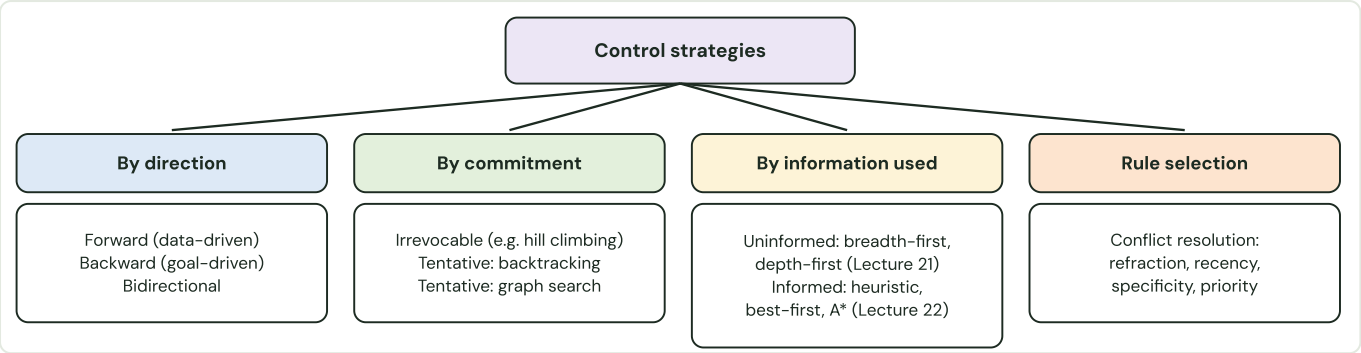
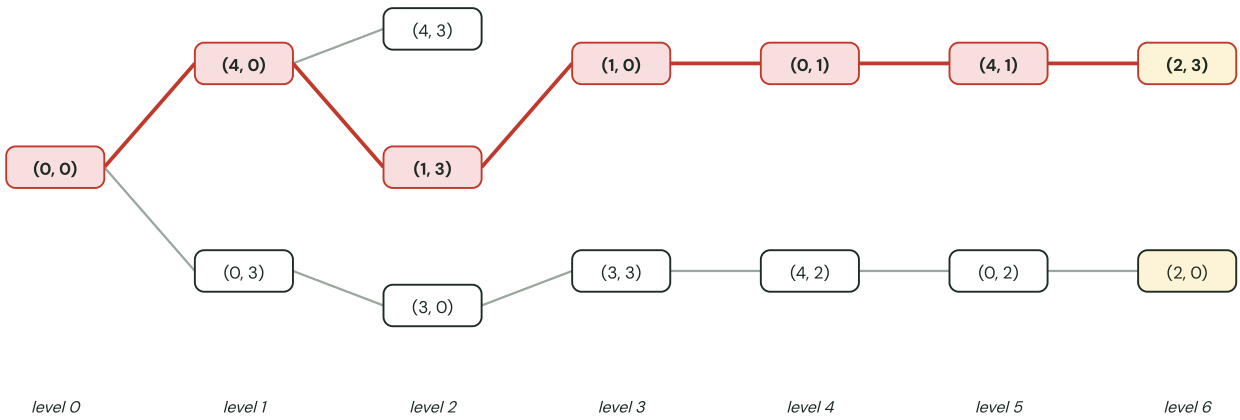


Figure 3. Ways of classifying control strategies. Original diagram.

Classification	Types	Explanation
Direction	Forward, backward, bidirectional	From the initial state towards the goal (data-driven), from the goal back (goal-driven), or both at once (Lecture 17)
Commitment	Irrevocable or tentative	An irrevocable strategy applies a rule and never reconsiders it (as in hill climbing); a tentative strategy keeps the possibility of returning — by **backtracking** to the last choice point, or by **graph search** that keeps a record of many alternatives (Nilsson, 1980)
Information used	Uninformed (blind) or informed (heuristic)	Blind strategies such as breadth-first and depth-first search use no knowledge of how close a state is to the goal (Lecture 21); heuristic strategies use an estimate to choose promising states first (Lecture 22)
Rule selection in rule-based systems	Conflict-resolution strategies	When several rules match, choose by refraction (do not fire the same rule on the same data twice), recency (prefer rules matching the newest facts), specificity (prefer more specific rules) or assigned priority

6. Worked example: measuring pesticide concentrate

Worked example 2 — the water-jug problem in a spray shed
A farmer must measure exactly 2 litres of concentrate using an unmarked 4-litre can and an unmarked 3-litre can, and may fill, empty or pour between the cans.
States: (x, y) = litres in the 4-L and 3-L cans; initial state (0, 0); goal: x = 2.
Rules: fill 4-L → (4, y); fill 3-L → (x, 3); empty 4-L → (0, y); empty 3-L → (x, 0); pour 4-L into 3-L until the 3-L can is full or the 4-L can is empty; pour 3-L into 4-L likewise.
Breadth-first control examines all states one step from the start, then all states two steps away, and so on, so the first goal it reaches is one of the shortest solutions.



States are (litres in 4-L can, litres in 3-L can). Red path: the breadth-first solution; yellow: goal states.

Figure 4. Breadth-first exploration of the water-jug state space, level by level. Original diagram generated from the same rules as the program.

Comparing breadth-first and depth-first control in Python (Lecture 8)

```
from collections import deque

A, B, GOAL = 4, 3, 2          # can sizes (litres) and the amount wanted in the 4-L can

def moves(state):             # production rules: each returns (rule name, new state)
    x, y = state
    yield "fill 4-L can", (A, y)
    yield "fill 3-L can", (x, B)
    yield "empty 4-L can", (0, y)
    yield "empty 3-L can", (x, 0)
    p = min(x, B - y); yield "pour 4-L into 3-L", (x - p, y + p)
    p = min(y, A - x); yield "pour 3-L into 4-L", (x + p, y - p)

def search(strategy):         # the control strategy decides which state to expand next
    frontier = deque([((0, 0), [])])
    seen, expanded = {(0, 0)}, 0
    while frontier:
        state, path = frontier.popleft() if strategy == "breadth-first" else frontier.pop()
        expanded += 1
        if state[0] == GOAL:
            return path, expanded
        for rule, new in moves(state):
            if new not in seen:            # systematic: never revisit a state
                seen.add(new)
                frontier.append((new, path + [(rule, new)]))
    return None, expanded

for strategy in ("breadth-first", "depth-first"):
    path, expanded = search(strategy)
    print(f"{strategy}: {len(path)} steps, {expanded} states expanded")
    for rule, state in path:
        print(f"   {rule:18s} -> (4-L: {state[0]}, 3-L: {state[1]})")

Output:
breadth-first: 6 steps, 13 states expanded
   fill 4-L can       -> (4-L: 4, 3-L: 0)
   pour 4-L into 3-L  -> (4-L: 1, 3-L: 3)
   empty 3-L can      -> (4-L: 1, 3-L: 0)
   pour 4-L into 3-L  -> (4-L: 0, 3-L: 1)
   fill 4-L can       -> (4-L: 4, 3-L: 1)
   pour 4-L into 3-L  -> (4-L: 2, 3-L: 3)
depth-first: 6 steps, 7 states expanded
   fill 3-L can       -> (4-L: 0, 3-L: 3)
   pour 3-L into 4-L  -> (4-L: 3, 3-L: 0)
   fill 3-L can       -> (4-L: 3, 3-L: 3)
   pour 3-L into 4-L  -> (4-L: 4, 3-L: 2)
   empty 4-L can      -> (4-L: 0, 3-L: 2)
   pour 3-L into 4-L  -> (4-L: 2, 3-L: 0)
```

The program was run and the output shown is its actual output. Both strategies found a 6-step solution here; depth-first control expanded fewer states (7 against 13) but, unlike breadth-first, it does not guarantee the fewest steps in general. Lecture 21 compares the two in detail.

The same ideas control agricultural AI systems: a weeding robot searching for a route between rows, a diagnosis program choosing which question to ask next (backward chaining, Lecture 17), or an irrigation planner choosing which rule to apply when several conditions hold.

7. Key takeaways

- Turing replaced "Can machines think?" with the imitation game: a machine passes if an interrogator, using typed questions, cannot reliably tell it from a person.
- Turing predicted that in about fifty years an average interrogator would have no more than a 70% chance of a correct identification after five minutes.
- He answered nine objections, including the theological, mathematical, consciousness and Lady Lovelace objections.
- Searle's Chinese room argues that behaviour is not understanding; ELIZA showed how easily people over-attribute understanding; CAPTCHAs reverse the test.
- Problems can be solved as searches through a state space defined by an initial state, operators, a goal test and path cost.
- A production system has a global database, production rules and a control strategy; a good strategy causes motion and is systematic.
- Control strategies differ in direction, commitment (irrevocable or tentative), information (blind or heuristic) and conflict resolution.

8. Examination questions

Short-answer questions (2 marks)

Q1. What question did Turing replace with the imitation game?

"Can machines think?"

Q2. Why are answers typewritten in the imitation game?

So that tones of voice cannot help the interrogator.

Q3. State Turing's 1950 prediction.

In about fifty years, an average interrogator would have no more than a 70% chance of identifying the machine after five minutes of questioning.

Q4. What is Lady Lovelace's objection, and how did Turing reply?

That a machine can only do what it is told and never surprise us; Turing said machines surprise him frequently.

Q5. What is the Chinese room argument?

Searle's argument that following rules to manipulate symbols, as a program does, is not by itself understanding.

Q6. What is a CAPTCHA?

A test that people pass easily but machines find hard, used to tell humans from programs — a reverse Turing test.

Q7. Name the components of a production system.

Global database, production rules and a control system.

Q8. State the two requirements of a good control strategy.

It must cause motion and be systematic.

Q9. Distinguish irrevocable and tentative control strategies.

Irrevocable strategies never reconsider an applied rule; tentative strategies can return to earlier choices by backtracking or graph search.

Q10. Define the state space of the water-jug problem.

States (x, y) of litres in the 4-L and 3-L cans, from (0, 0), with fill, empty and pour operators, goal x = 2.

Long-answer questions (5–10 marks) — answer outlines

Q1. Describe the Turing test with a diagram and discuss its significance.

Imitation game, Figure 1, typewritten answers, machine takes A's part; prediction; worked example 1.

Q2. Discuss the objections Turing considered and the later criticisms of the test.

Nine objections table; Chinese room, ELIZA, CAPTCHA, total Turing test.

Q3. Explain state-space representation and production systems.

States, initial state, operators, goal test, path cost; Figure 2.

Q4. Explain control strategies and their requirements and classification.

Motion and systematic; Figure 3 and table — direction, commitment, information, conflict resolution.

Q5. Solve the water-jug problem by state-space search.

Worked example 2; Figure 4; Python program and both solutions.

Q6. Compare breadth-first and depth-first control using an example.

Program output — steps and states expanded; guarantees; link to Lecture 21.

9. Glossary

Term	Meaning
Backtracking	Returning to the most recent choice point to try another rule
CAPTCHA	A test that tells humans from machines — a reverse Turing test
Chinese room	Searle's argument that symbol manipulation is not understanding
Conflict resolution	Choosing among several rules that match at the same time
Control strategy	The procedure that decides which rule or state to try next
Goal test	Check of whether a state solves the problem
Graph search	A tentative strategy that keeps a record of many explored alternatives
Imitation game	Turing's test in which an interrogator tries to tell a machine from a person
Irrevocable strategy	A strategy that never reconsiders a rule once applied
Operator	An action or rule that changes one state into another
Production rule	An IF condition THEN action rule
Production system	Global database + production rules + control system
State space	All states reachable from the initial state by the operators
Turing test	Whether a machine can make its conversation indistinguishable from a person's

References

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All diagrams and the program are original. Quotations from Turing (1950) are short extracts for teaching; the summaries of the objections are paraphrased. Data in worked example 1 are illustrative.

Practice MCQs

Q1. Turing proposed the imitation game to replace the question

(a) "Can machines calculate?" (b) "Can machines think?"
(c) "Can machines see?" (d) "Can machines feel pain?"

Q2. In the imitation game the answers are typewritten so that

(a) they are faster (b) tones of voice cannot help the interrogator
(c) the machine can read them (d) they can be printed

Q3. Turing predicted an average interrogator would have no more than what chance of a correct identification after five minutes?

(a) 50% (b) 70%
(c) 90% (d) 100%

Q4. "Machines take me by surprise with great frequency" was Turing's reply to

(a) the theological objection (b) Lady Lovelace's objection
(c) the mathematical objection (d) the ESP objection

Q5. Searle's Chinese room argument claims that

(a) machines will soon pass the Turing test (b) symbol manipulation alone is not understanding
(c) Chinese is hard to translate (d) computers cannot store text

Q6. A CAPTCHA is best described as

(a) a type of computer virus (b) a reverse Turing test
(c) a search algorithm (d) a database

Q7. ELIZA (1966) produced replies by

(a) understanding meaning (b) keywords and reply templates
(c) searching the Internet (d) deep learning

Q8. The component of a production system that decides which rule to apply next is the

(a) global database (b) control strategy
(c) rule set (d) printer

Q9. The two requirements of a good control strategy are that it

(a) is fast and cheap (b) causes motion and is systematic
(c) uses heuristics and graphs (d) is forward and backward

Q10. A strategy that never reconsiders a rule once applied is

(a) tentative (b) irrevocable
(c) backtracking (d) graph search

Q11. In the water-jug problem with 4-L and 3-L cans, pouring the full 4-L can into the empty 3-L can gives

(a) (0, 3) (b) (1, 3)
(c) (4, 0) (d) (3, 1)

Q12. Breadth-first control guarantees

(a) the fewest states expanded (b) a solution with the fewest steps
(c) no memory use (d) random order

Answers

Q1 — (b) "Can machines think?"
Q2 — (b) tones of voice cannot help the interrogator
Q3 — (b) 70%
Q4 — (b) Lady Lovelace's objection
Q5 — (b) symbol manipulation alone is not understanding
Q6 — (b) a reverse Turing test
Q7 — (b) keywords and reply templates
Q8 — (b) control strategy
Q9 — (b) causes motion and is systematic
Q10 — (b) irrevocable
Q11 — (b) (1, 3)
Q12 — (b) a solution with the fewest steps